



Artificial Intelligence Foundations (CIS4049-N)

Week 2 – Intelligent Agents – Part 2

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Acting Rationally – “Doing the right thing”

- The “right thing”: **maximize utility** given available information
- Does not necessarily require “thought” (e.g., reflexes)
- **Advantages** of AI as design of **rational agents**:
 - **more general** than “laws of thought”: logical inference only one mechanism for obtaining rational behavior
 - better suited for **scientific method** than approaches based on acting/thinking humanly

Rational Agents

- AI systems – very different tasks to solve
- How do we capture this diversity in a systematic framework emphasizing **commonalities** and **differences**?
- Common metaphor: **rational agents** and **their environments**

Rational Decisions

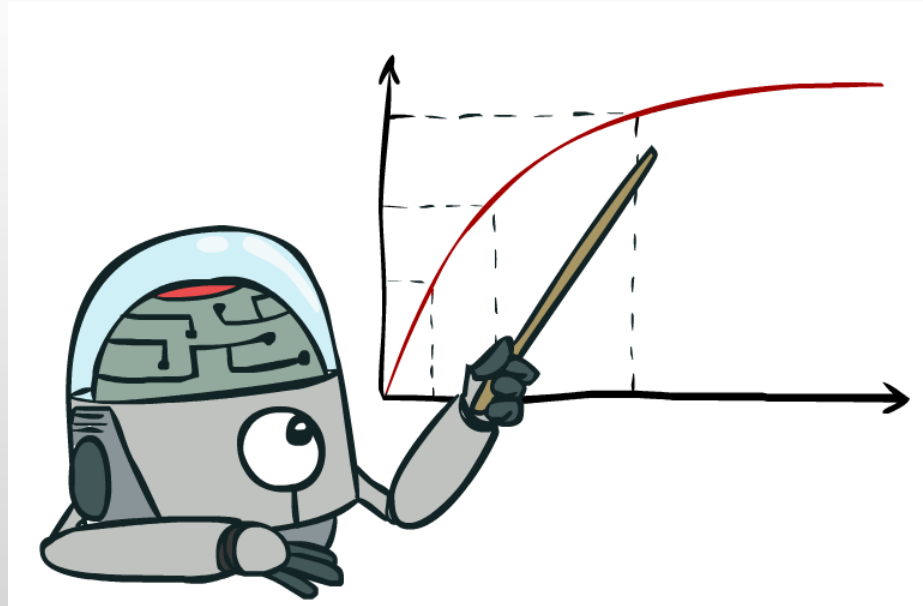
We'll use the term **rational** in a very specific, technical way:

- Rational: maximally achieving pre-defined goals
- Rationality only concerns what decisions are made
- (not the thought process behind them)
- Goals are expressed in terms of the **utility** of outcomes
- Being rational means **maximizing your expected utility**

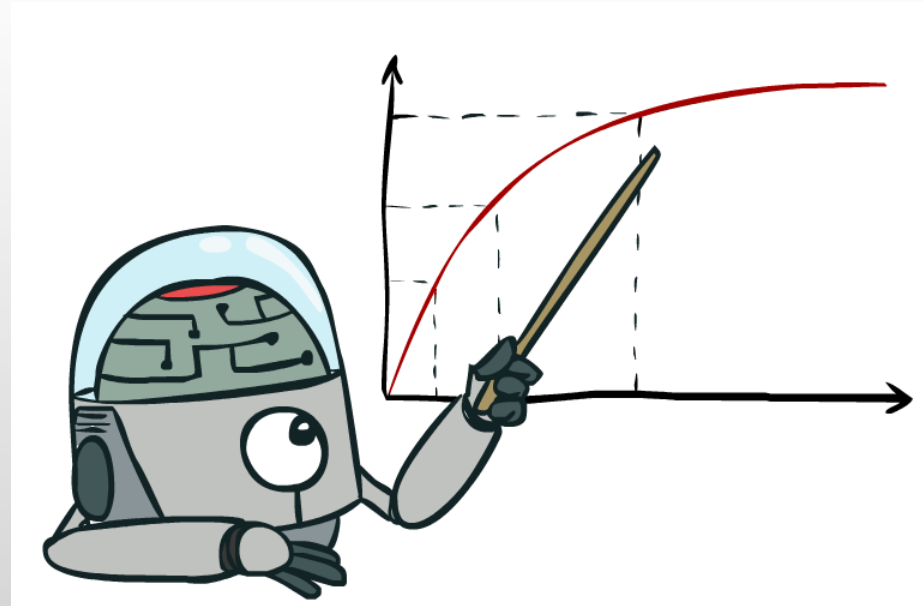


Computational Rationality

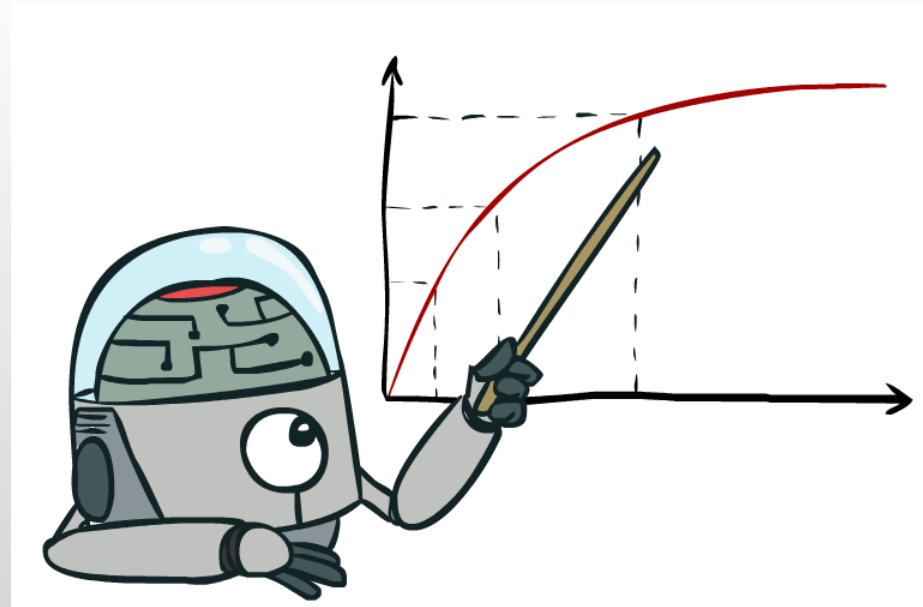
Maximize Your Expected Utility



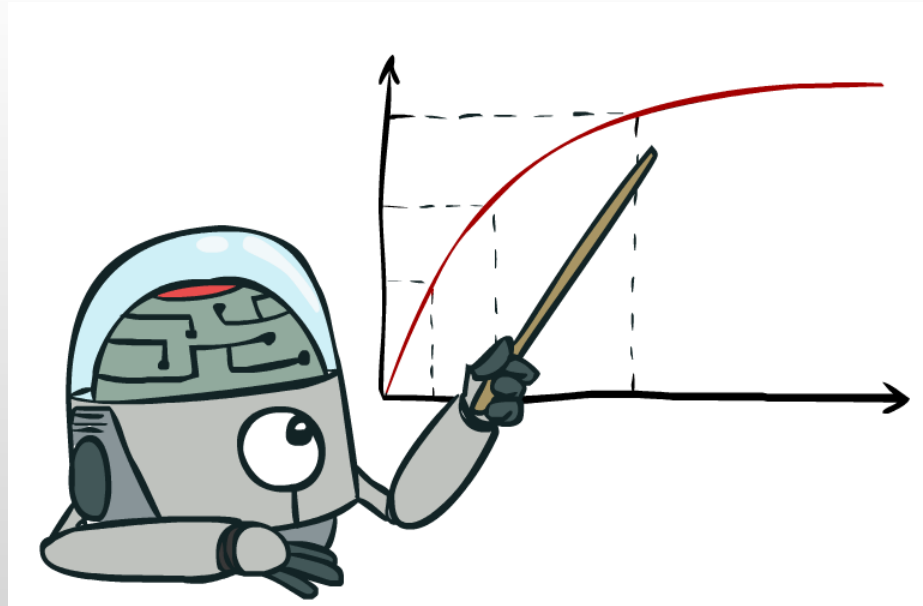
Maximise Your Expected Utility



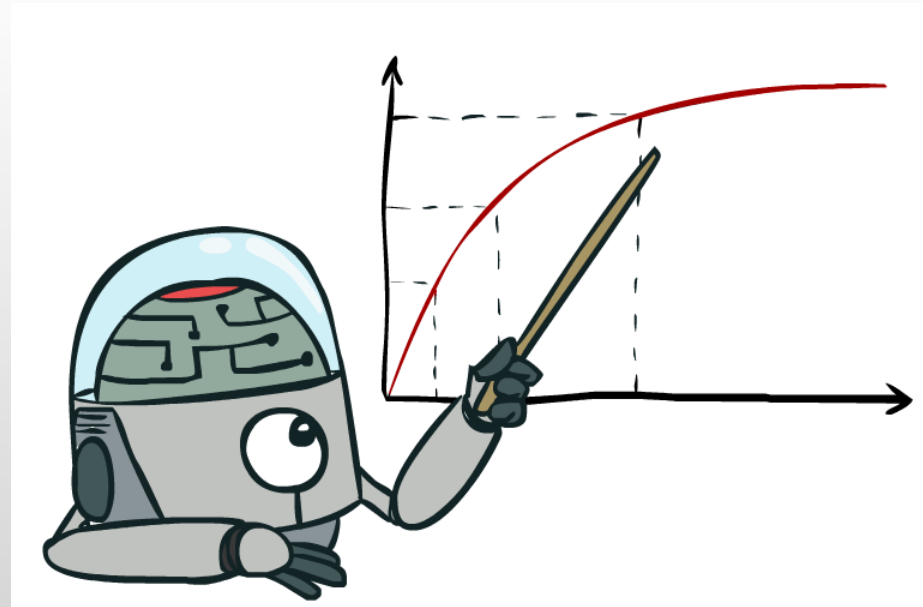
Maximise Your Expected Utility



Maximise Your Expected Utility

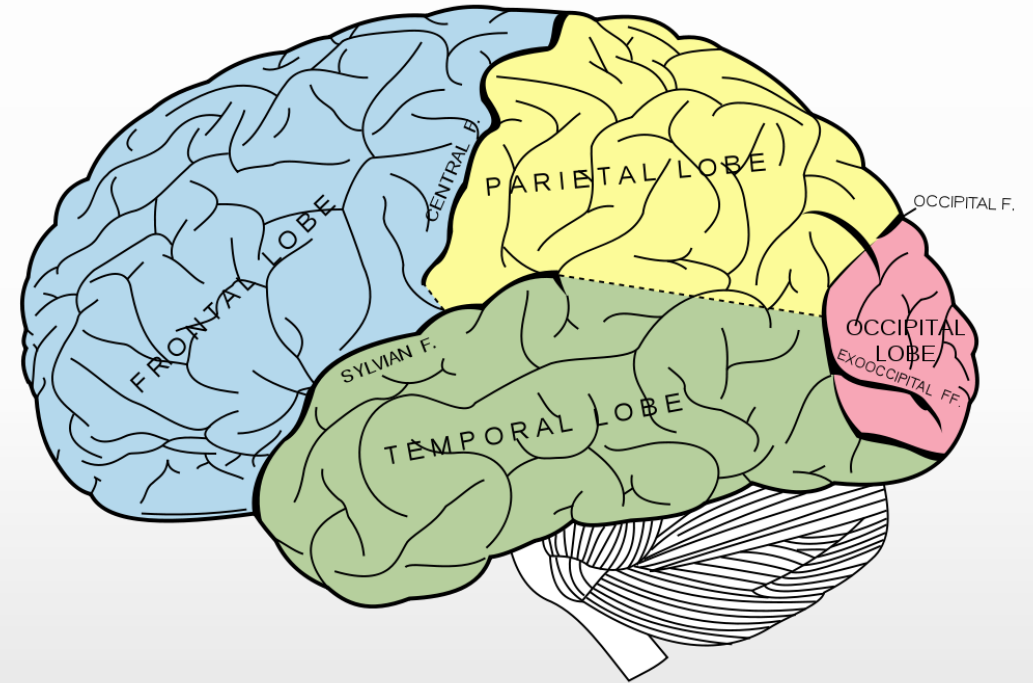


Maximise Your Expected Utility



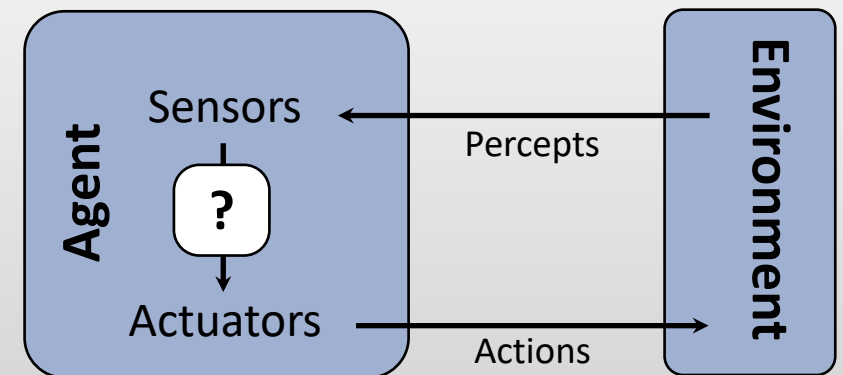
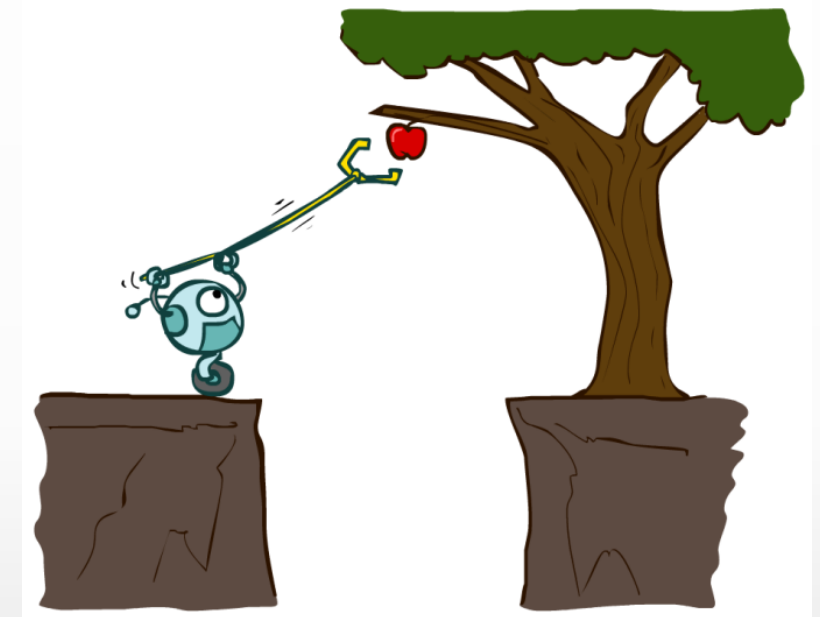
What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- “Brains are to intelligence as wings are to flight”
- Lessons learned from the brain: memory and simulation are key to decision making

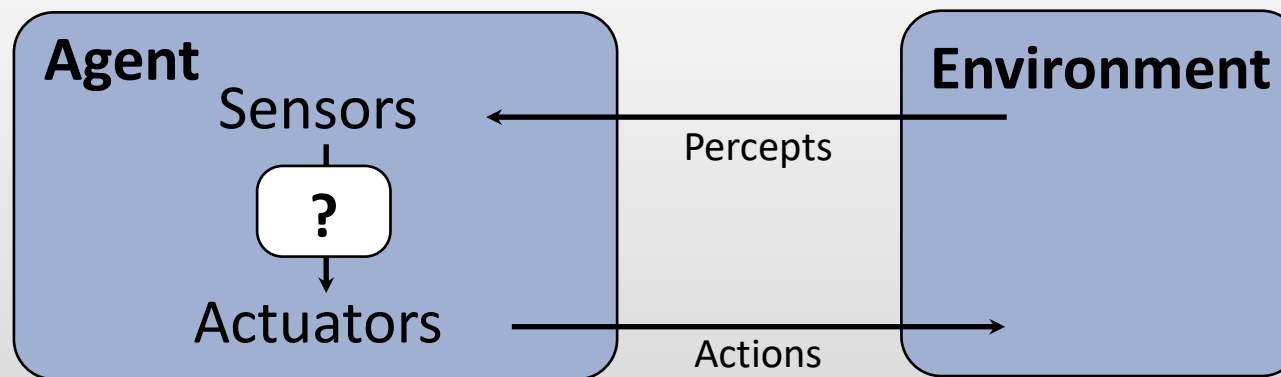
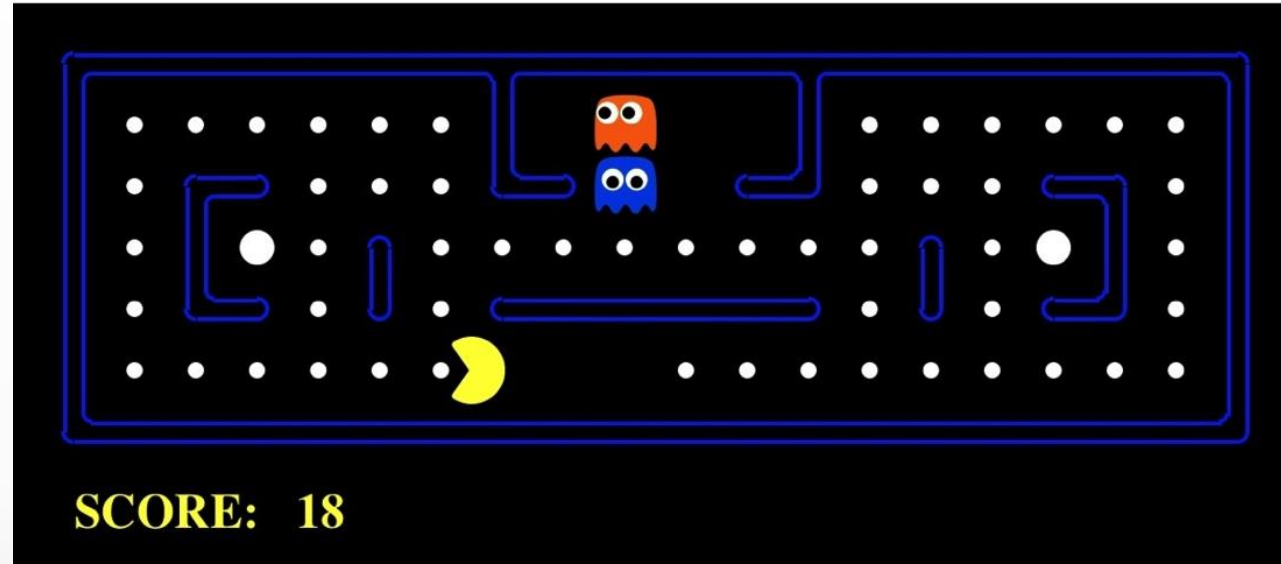


Rational Agents

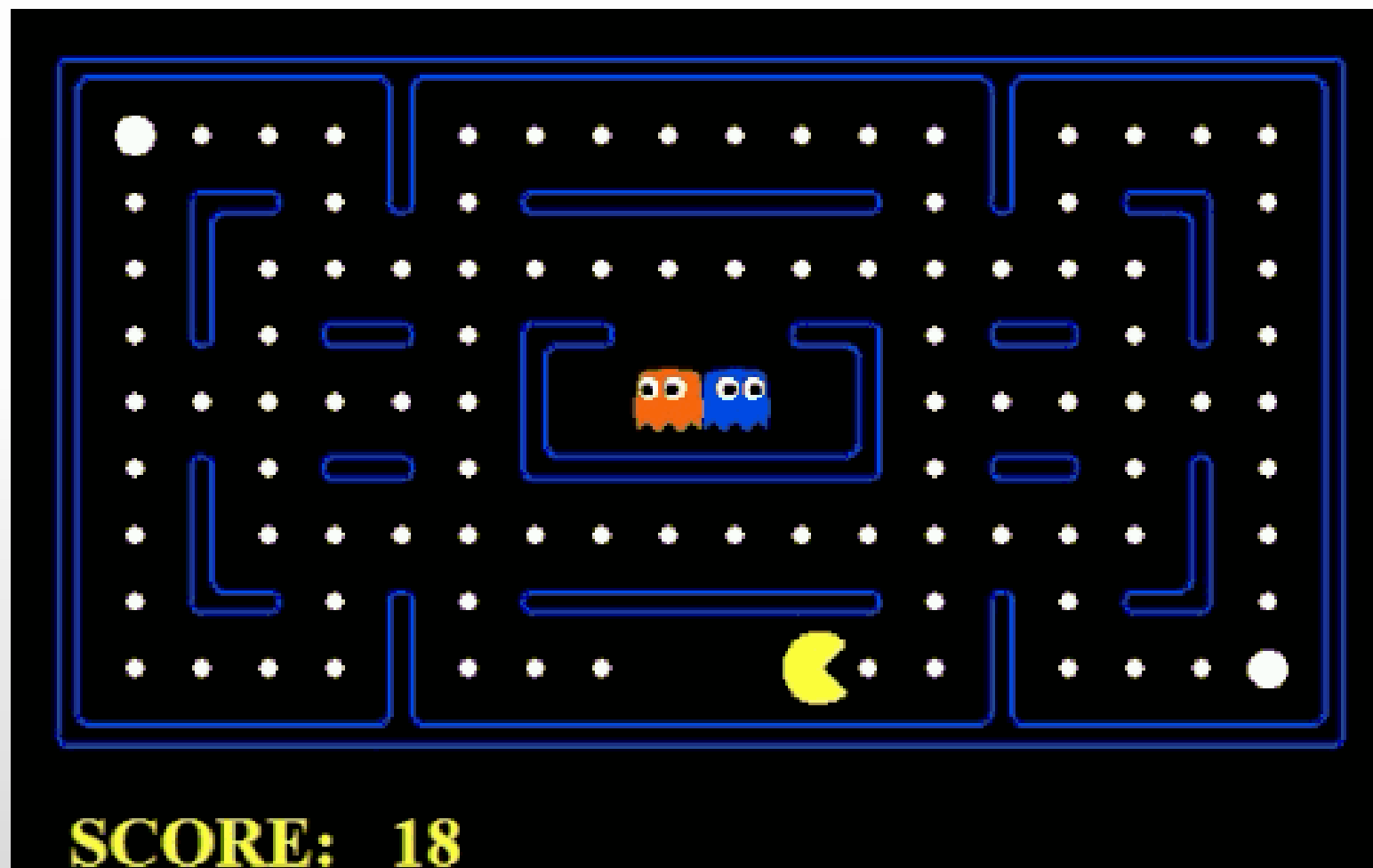
- An **agent** is an entity that *perceives* and *acts*
- A **rational agent** selects actions that maximize its (expected) **utility**
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions
- This **module** is about:
 - General AI techniques for a variety of problem types
 - Learning to recognize when and how a new problem can be solved with an existing technique

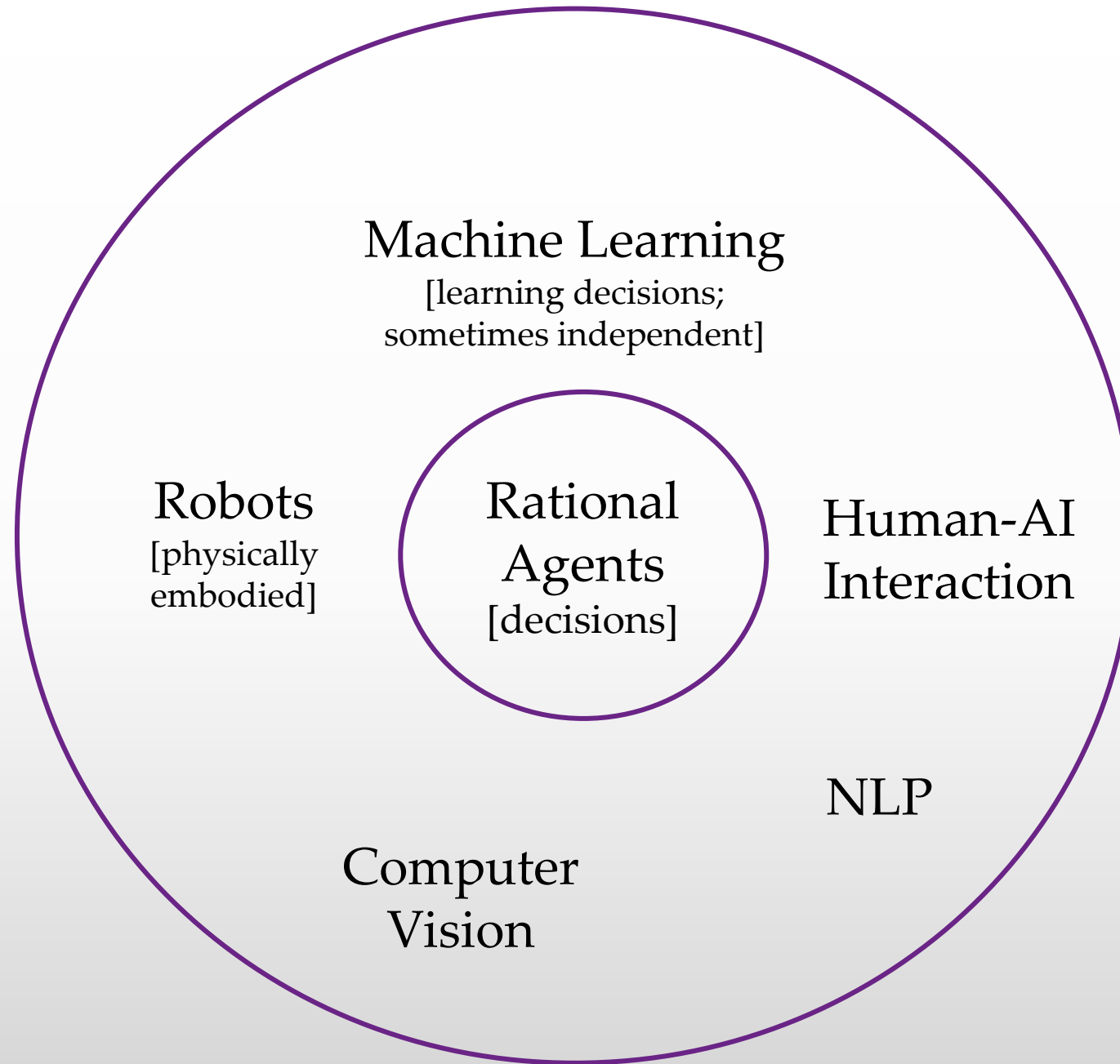


Pac-Man as an Agent

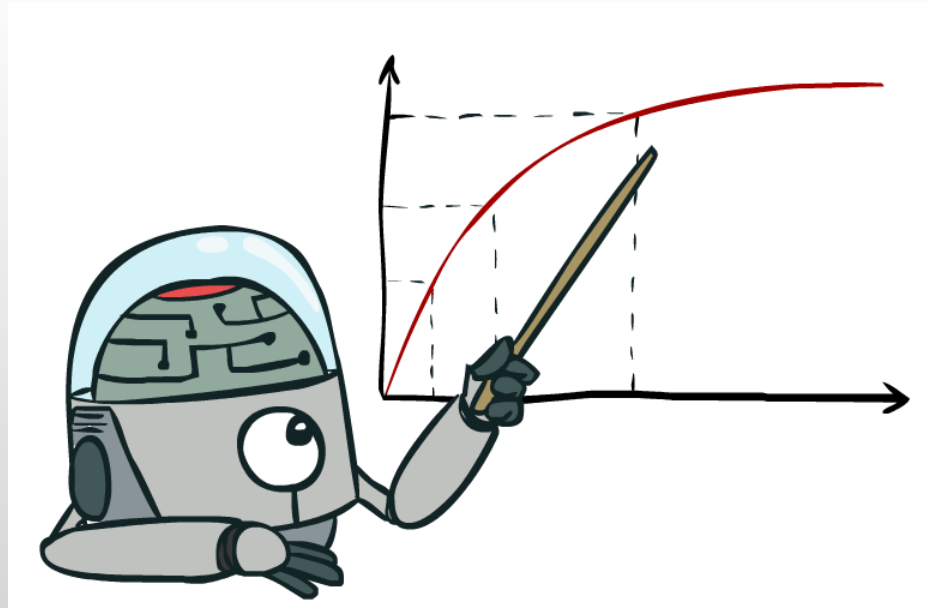


Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes





Maximize Your Expected Utility

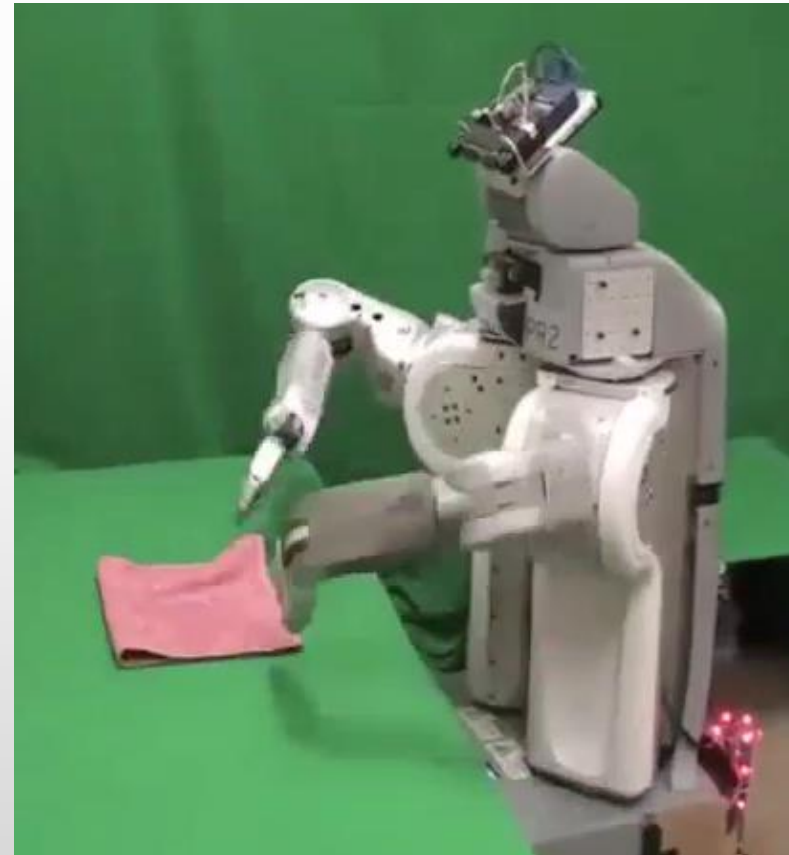


Utility?

Clear utility function



Not so clear utility function





Maximize Your Expected Utility

